

Electric Charge and Electric Field

The force that holds atoms together: electric charge, Coulomb's law, and the field that carries the force through space.

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Every chemical bond, every nerve impulse, the friction under your shoes and the stiffness of steel are, at bottom, electrical. The electric force is about 10^{39} times stronger than gravity. We rarely notice it only because matter is almost perfectly neutral, with positive and negative charges cancelling. This chapter opens the second semester. It introduces electric charge and Coulomb's law, then the central idea of electromagnetism: the **field**, a property of space created by charges that tells any other charge what force it will feel. We calculate fields for point charges and for continuous distributions, follow charged particles moving through fields, and study the electric dipole, the model behind water molecules, microwave ovens and the electrocardiogram.

YOU WILL BE ABLE TO

- Describe the properties of electric charge (quantised, conserved) and how objects become charged.
- Apply Coulomb's law with vector superposition.
- Define the electric field, draw field lines, and calculate the fields of point charges.
- Calculate the fields of continuous charge distributions (rod, ring, disc, line, sheet) by integration.
- Analyse the motion of charged particles in uniform fields.
- Describe electric dipoles: their dipole moment, field, and the torque and energy in an external field.

THINK FIRST

Rub a balloon on your hair and it sticks to a wall. The wall is electrically neutral, so why is it attracted to the charged balloon at all?

The balloon's charge **polarises** the wall. If the balloon is negative, it pushes the electrons in the wall's molecules slightly away and pulls their positive nuclei slightly closer. The wall stays neutral overall, but its positive side now faces the balloon, a little *nearer* than the negative side. Coulomb's force weakens with distance, so the attraction to the nearer positive charges wins over the repulsion from the farther negative ones. The net force is attractive. The same *induced-dipole* effect explains how charged dust clings to screens and how some molecules stick together.

§13.1

Electric charge

Charge is a basic property of matter, measured in **coulombs** (C).

- There are **two kinds**, positive and negative. Like charges repel and unlike charges attract.
- Charge is **quantised**: every free charge is a whole multiple of the elementary charge, $e = 1.602 \times 10^{-19}$ C. A proton has charge $+e$ and an electron $-e$. (Quarks carry $\pm \frac{1}{3}e$ and $\pm \frac{2}{3}e$, but they are never found alone.)
- Charge is **conserved**: the net charge of an isolated system never changes. Rubbing two objects together transfers electrons from one to the other. It does not create charge.

Conductors and insulators. In **conductors** (metals, salt solutions, body fluids) some charges are free to move. In metals these are electrons, and in electrolytes they are ions. In **insulators** (glass, plastic, dry skin) charges are bound to their atoms. **Semiconductors** lie in between, and their conductivity can be engineered (Chapter 24).

Charging by induction. Bring a charged rod near a neutral conductor. The conductor's free electrons redistribute, leaving one side positive and the other negative. Now earth the far side briefly with a wire, then remove the earth connection before taking the rod away. The conductor is left with a net charge *opposite* to the rod's, and the rod has not touched it.

§13.2

Coulomb's law

Two point charges q_1 and q_2 a distance r apart exert forces on each other along the line joining them, of magnitude

$$F = k \frac{|q_1 q_2|}{r^2}, \quad k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2,$$

where $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$ is the **permittivity of free space**. In vector form, the force on charge 2 due to charge 1 is

$$\vec{F}_{12} = k \frac{q_1 q_2}{r^2} \hat{r}_{12},$$

where $\hat{r}_{12}$ points from charge 1 to charge 2. A positive product $q_1 q_2$ means repulsion.

Coulomb's law has the same inverse-square form as gravitation, but charges come in two signs and the force is enormously stronger. Forces from several charges add as vectors: this is **superposition**.

WORKED EXAMPLE 13.1

Electric versus gravitational force in hydrogen

In a hydrogen atom, the electron and proton are on average $5.29 \times 10^{-11} \text{ m}$ apart. Compare the electric and gravitational forces between them.

Solution

$$F_e = \frac{(8.99 \times 10^9)(1.602 \times 10^{-19})^2}{(5.29 \times 10^{-11})^2} = 8.2 \times 10^{-8} \text{ N}.$$

$$F_g = \frac{(6.67 \times 10^{-11})(9.11 \times 10^{-31})(1.67 \times 10^{-27})}{(5.29 \times 10^{-11})^2} = 3.6 \times 10^{-47} \text{ N}.$$

$$\frac{F_e}{F_g} = 2.3 \times 10^{39}.$$

Evaluate. Gravity is utterly negligible in atoms, molecules, chemistry and biology. It dominates on astronomical scales only because large bodies are electrically neutral, so their huge electric forces cancel, while gravity, which always attracts, adds up.

WORKED EXAMPLE 13.2

Superposition of forces

A charge $q_1 = +2.0 \mu\text{C}$ sits at the origin and $q_2 = -3.0 \mu\text{C}$ at $(0.30, 0)$ m. Find the net force on $q_3 = +1.0 \mu\text{C}$ at $(0, 0.40)$ m.

Solution

Force from q_1 (repulsive, along $+y$, at a distance of 0.40 m):

$$F_{13} = \frac{(8.99 \times 10^9)(2.0 \times 10^{-6})(1.0 \times 10^{-6})}{0.16} = 0.112 \text{ N}, \quad \vec{F}_{13} = 0.112 \hat{j} \text{ N}.$$

Force from q_2 (attractive, so it points from q_3 towards q_2 , at a distance of 0.50 m):

$$F_{23} = \frac{(8.99 \times 10^9)(3.0 \times 10^{-6})(1.0 \times 10^{-6})}{0.25} = 0.108 \text{ N}.$$

The unit vector from q_3 towards q_2 is $(0.30, -0.40)/0.50 = (0.60, -0.80)$, so

$$\vec{F}_{23} = 0.0647 \hat{i} - 0.0863 \hat{j} \text{ N}.$$

Net force:

$$\vec{F} = 0.0647 \hat{i} + 0.0261 \hat{j} \text{ N}, \quad F = 0.070 \text{ N at } \tan^{-1} \frac{0.0261}{0.0647} = 22^\circ \text{ above the } +x \text{ axis}.$$

§13.3

The electric field

Instead of saying “charge A exerts a force on charge B”, we say that charge A creates an **electric field** $\vec{E}$ throughout the space around it, and charge B responds to the field where it sits. The field at a point is the force per unit charge on a small positive test charge q_0 placed there:

$$\vec{E} = \frac{\vec{F}}{q_0}, \quad \vec{F} = q\vec{E}.$$

The units are N/C, which equal V/m (Chapter 15). A positive charge feels a force along $\vec{E}$, and a negative charge feels a force opposite to it.

The field of a **point charge** q at distance r is

$$\vec{E} = k \frac{q}{r^2} \hat{r}.$$

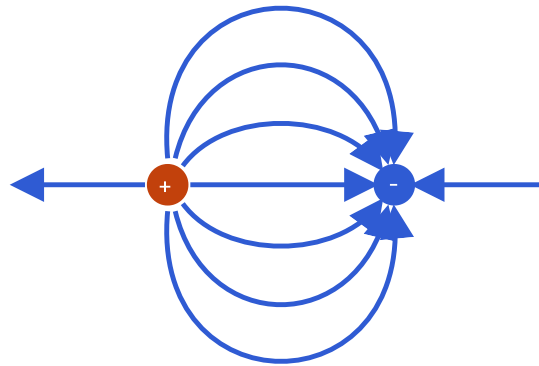
It points directly away from a positive charge and directly towards a negative one. Fields from several sources add as vectors.

The field is more than a calculating device. When a charge moves, the change in its field spreads outward at the speed of light (Chapter 21). The field carries energy and momentum, so it is physically real.

Field lines

Field lines are a picture of the field. They follow these rules:

- The tangent to a field line gives the direction of $\vec{E}$.
- The lines are crowded together where the field is strong.
- Lines **begin on positive charges and end on negative charges** (or at infinity).
- Field lines **never cross**, because the field has only one direction at each point.



Field lines of an electric dipole. They leave the positive charge and end on the negative charge, crowding together where the field is strongest.

WORKED EXAMPLE 13.3

Where is the field zero?

A charge of $+4.0 \mu\text{C}$ sits at $x = 0$ and a charge of $+1.0 \mu\text{C}$ at $x = 0.30 \text{ m}$. Where on the x -axis is the electric field zero?

Solution

The two fields can cancel only *between* the charges, where they point in opposite directions. Outside, both point the same way. At position x :

$$\frac{k(4.0 \mu\text{C})}{x^2} = \frac{k(1.0 \mu\text{C})}{(0.30 - x)^2} \Rightarrow \frac{2}{x} = \frac{1}{0.30 - x} \Rightarrow x = 0.20 \text{ m.}$$

The zero is closer to the smaller charge, as it must be.

§13.4

Fields of continuous charge distributions

For charge spread over an object, split it into small elements dq . Each element contributes a field $d\vec{E} = k dq \hat{r}/r^2$, and the total field is the integral of these contributions. Use the **charge density** that fits the shape:

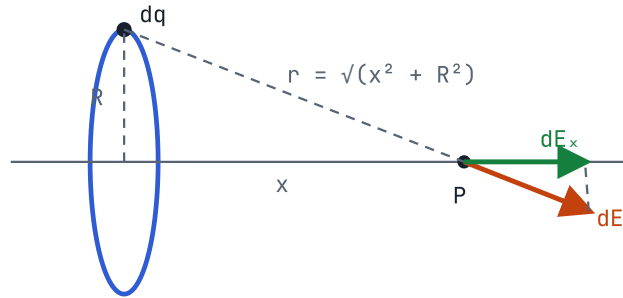
- $dq = \lambda dx$ for a line (λ in C/m);
- $dq = \sigma dA$ for a surface (σ in C/m²);
- $dq = \rho dV$ for a volume (ρ in C/m³).

KEY IDEA

Strategy for continuous distributions.

1. Draw a typical element dq and the field $d\vec{E}$ it produces at the field point.
2. Use **symmetry** to decide which components of $d\vec{E}$ cancel.
3. Integrate only the component that survives.

Uniformly charged ring (charge Q , radius R), at a point on its axis a distance x from the centre:



Field on the axis of a charged ring. By symmetry, the perpendicular components from opposite sides of the ring cancel, leaving only the axial components.

Every element is the same distance $r = \sqrt{x^2 + R^2}$ from P. The components perpendicular to the axis cancel in pairs. The axial component of each contribution is $dE \cos \alpha$, with $\cos \alpha = x/r$:

$$E_x = \int \frac{k dq}{r^2} \cdot \frac{x}{r} = \frac{kx}{(x^2 + R^2)^{3/2}} \int dq = \frac{kQx}{(x^2 + R^2)^{3/2}}.$$

Checks. $E = 0$ at the centre ($x = 0$), by symmetry. For $x \gg R$, $E \rightarrow kQ/x^2$, so from far away the ring looks like a point charge. The field is largest at $x = R/\sqrt{2}$.

Uniformly charged disc (charge density σ , radius R), on its axis. Build the disc from rings of radius r' , width dr' and charge $dq = \sigma 2\pi r' dr'$, and add up the ring fields:

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right).$$

Infinite sheet (let $R \rightarrow \infty$):

$$E = \frac{\sigma}{2\epsilon_0},$$

which is **uniform**: the same at every distance from the sheet.

Infinite line of charge (linear density λ), at perpendicular distance r :

$$E = \frac{2k\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r}.$$

Problem P13.13 treats a finite rod. Chapter 14 obtains the line and sheet results in a few lines using Gauss's law.

WORKED EXAMPLE 13.4

The field of a charged ring

A ring of radius 5.0 cm carries 50 nC spread uniformly. Find the field on its axis at 12 cm from the centre, and the largest field anywhere on the axis.

Solution

At $x = 0.12$ m:

$$E = \frac{(8.99 \times 10^9)(5.0 \times 10^{-8})(0.12)}{(0.12^2 + 0.05^2)^{3/2}} = \frac{53.9}{(0.0169)^{3/2}} = \frac{53.9}{2.197 \times 10^{-3}} = 2.5 \times 10^4 \text{ N/C.}$$

The maximum is at $x = R/\sqrt{2} = 0.0354$ m:

$$E_{\max} = \frac{kQ(R/\sqrt{2})}{(1.5R^2)^{3/2}} = \frac{2kQ}{3\sqrt{3}R^2} = \frac{2(449.5)}{3\sqrt{3}(0.0025)} = 6.9 \times 10^4 \text{ N/C.}$$

§13.5

Charged particles in uniform fields

A uniform field gives a particle of charge q and mass m a constant acceleration

$$\vec{a} = \frac{q\vec{E}}{m}.$$

Everything from Chapter 2 then applies. A particle entering the field at right angles follows a **parabola**, just like a projectile under gravity. Because electrons and ions have very large charge-to-mass ratios, even modest fields produce enormous accelerations.

A uniform field can be produced between two large parallel plates with equal and opposite charges, where $E = \sigma/\epsilon_0$ (Chapter 14).

WORKED EXAMPLE 13.5

Deflecting an electron beam

An electron moving horizontally at 2.0×10^7 m/s enters the space between two parallel plates 5.0 cm long, where there is a uniform downward field of 2.0×10^4 N/C. Find its vertical deflection on leaving the plates. Gravity is negligible.

Solution

The force on the electron is upward, opposite to $\vec{E}$, because its charge is negative:

$$a = \frac{eE}{m} = \frac{(1.602 \times 10^{-19})(2.0 \times 10^4)}{9.11 \times 10^{-31}} = 3.5 \times 10^{15} \text{ m/s}^2.$$

Time between the plates:

$$t = \frac{0.050}{2.0 \times 10^7} = 2.5 \times 10^{-9} \text{ s}.$$

Deflection:

$$y = \frac{1}{2}at^2 = \frac{1}{2}(3.5 \times 10^{15})(2.5 \times 10^{-9})^2 = 0.011 \text{ m} = 1.1 \text{ cm upward}.$$

Evaluate. The acceleration is about $10^{14}g$, so ignoring gravity is fully justified. This is how the beam in an old cathode-ray tube or oscilloscope was steered. The same principle now steers ions in mass spectrometers and drops of ink in industrial inkjet printers.

IN PRACTICE

Separating molecules with electric fields. In gel electrophoresis, DNA fragments, which are negatively charged, are pulled through a gel by an applied field of about 100 V/m. They quickly reach a terminal velocity at which the electric force qE balances the drag force. Smaller fragments slip through the gel's pores more easily, so they travel farther in a given time, sorting the fragments by size. This is the basis of DNA fingerprinting, PCR product checks and genetic testing. Protein electrophoresis of blood serum separates albumin and the globulins, helping diagnose conditions such as multiple myeloma. Electrostatic precipitators in power-station chimneys use the same physics: they charge smoke particles and drive them onto collecting plates, removing over 99% of the particulates.

§13.6

Electric dipoles

An **electric dipole** is a pair of equal and opposite charges $\pm q$ separated by a distance d . Its **dipole moment** is the vector

$$\vec{p} = q\vec{d},$$

which points from $-q$ to $+q$ and is measured in C m. Many molecules are permanent dipoles.

Water, for example, has $p = 6.2 \times 10^{-30}$ C m because oxygen pulls electron density away from the hydrogens. Others are *induced* dipoles, as in the balloon of the “Think first” box.

Field of a dipole far away ($r \gg d$):

- on the axis: $E = \frac{2kp}{r^3}$;
- on the perpendicular bisector: $E = \frac{kp}{r^3}$, pointing antiparallel to $\vec{p}$.

The field falls off as $1/r^3$, faster than a point charge's $1/r^2$. From far away, the two opposite charges nearly cancel.

A dipole in a uniform external field feels zero net force, because $q\vec{E} - q\vec{E} = 0$, but it does feel a torque that tries to align it with the field:

$$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin \theta.$$

Its potential energy is

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta.$$

This is lowest when $\vec{p}$ is aligned with $\vec{E}$ (stable) and highest when it points opposite to $\vec{E}$ (unstable). In a *non-uniform* field, a dipole also feels a net force, towards the region of stronger field.

WORKED EXAMPLE 13.6

Water molecules in a field

A water molecule ($p = 6.2 \times 10^{-30}$ C m) is in a field of 1.0×10^5 N/C. Find (a) the maximum torque on it and (b) the energy needed to turn it from aligned with the field to anti-aligned. Compare (b) with the typical thermal energy kT at 300 K.

Solution

(a) $\tau_{\max} = pE = (6.2 \times 10^{-30})(1.0 \times 10^5) = 6.2 \times 10^{-25}$ N m.

(b) $\Delta U = (+pE) - (-pE) = 2pE = 1.2 \times 10^{-24}$ J. The thermal energy is $kT = (1.38 \times 10^{-23})(300) = 4.1 \times 10^{-21}$ J.

Evaluate. Thermal agitation is about 3000 times stronger, so in such a field water molecules are only very slightly aligned on average. In a microwave oven, the field reverses 2.45 billion times a second. Molecules try to follow it, jostle their neighbours, and turn the field's energy into heat. Near an ion in solution, however, the field is about 10^{10}

N/C. There water molecules *are* strongly aligned, forming the hydration shells that control how ions behave in living cells.

IN PRACTICE

The heart as a dipole: the electrocardiogram (ECG). As a wave of electrical activity (depolarisation) spreads through the heart muscle, the boundary between excited and resting tissue acts like a moving electric dipole. Its strength and direction change throughout each heartbeat. Its field reaches the body surface, where electrodes record tiny potential differences of about 1 mV. The standard limb leads look at this “cardiac dipole” from different directions, like viewing an arrow from several angles. That is why the shape of the trace differs between leads, and why changes in particular leads point to damage in particular regions of the heart.

CHAPTER SUMMARY

- Charge is quantised ($e = 1.602 \times 10^{-19}$ C) and conserved. Conductors have mobile charges and insulators do not.
- Coulomb: $F = k|q_1q_2|/r^2$, with $k = 1/4\pi\epsilon_0 = 8.99 \times 10^9$ N m²/C². Forces add as vectors.
- Field: $\vec{E} = \vec{F}/q_0$, so $\vec{F} = q\vec{E}$. Point charge: $E = kq/r^2$, radially outward for positive q .
- Field lines start on +, end on −, never cross, and crowd together where the field is strong.
- Ring on its axis: $E = kQx/(x^2 + R^2)^{3/2}$. Disc: $E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}}\right)$. Infinite sheet: $\sigma/2\epsilon_0$. Infinite line: $\lambda/(2\pi\epsilon_0 r)$.
- A uniform field gives constant acceleration $q\vec{E}/m$, and a particle entering it sideways follows a parabola.
- Dipole: $\vec{p} = q\vec{d}$, field $\propto 1/r^3$, torque $\vec{\tau} = \vec{p} \times \vec{E}$ and energy $U = -\vec{p} \cdot \vec{E}$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P13.1

A rubbed balloon sticks to a neutral wall, and a charged comb picks up small, neutral bits of paper. Explain both in terms of polarisation. Would the effect work if the comb had the opposite charge?

P13.2

Why can electric field lines never cross? What would it mean if they did?

P13.3

A positive point charge is brought near an isolated, uncharged metal sphere. Is there a net force on the sphere? If so, in which direction? Explain.

P13.4

Explain qualitatively why the field of a dipole falls off as $1/r^3$ at large distances, rather than as $1/r^2$.

LEVEL B – STANDARD PROBLEMS

P13.5

Two small identical spheres, each of mass 0.10 g, hang from a common point on insulating threads 50 cm long. When given equal charges, they repel and each thread hangs at 10° to the vertical. Find the charge on each sphere.

P13.6

Four equal charges of $+1.0 \mu\text{C}$ sit at the corners of a square of side 10 cm. (a) What is the field at the centre of the square? (b) Find the magnitude and direction of the net force on any one of the charges.

P13.7

A charge of $+5.0 \text{ nC}$ sits at $(-5.0, 0) \text{ cm}$ and a charge of -5.0 nC at $(+5.0, 0) \text{ cm}$. Find the electric field (magnitude and direction) at the point $(0, 12) \text{ cm}$.

P13.8

An electron is released from rest in a uniform field of 500 N/C. (a) Find its speed after it has moved 2.0 cm, and the time this takes. (b) Repeat for a proton, and explain the difference.

P13.9

A ring of radius 5.0 cm carries 50 nC spread uniformly. Find the field on its axis 12 cm from the centre, and compare it with the field of a 50 nC point charge at the same distance.

P13.10

A long straight wire carries a uniform charge of $2.0 \mu\text{C}$ per metre. Find the electric field 10 cm from the wire. What force would act on an electron at that point?

P13.11

A dipole consists of charges of ± 2.0 nC separated by 4.0 mm. It is placed in a uniform field of 5.0×10^4 N/C, with its dipole moment at 30° to the field. Find (a) the dipole moment, (b) the torque, and (c) the work needed to turn it from aligned with the field to anti-aligned.

P13.12

Electrophoresis. A protein carries a net charge of $+10e$ and can be modelled as a sphere of radius 3.0 nm. It moves through a buffer solution of viscosity 1.0×10^{-3} Pa s in a field of 1000 N/C (1000 V/m). Using Stokes drag ($6\pi\eta r v$), find its terminal speed in cm per hour. How would the speed change for a protein of the same charge but twice the radius?

LEVEL C – CHALLENGE PROBLEMS**P13.13**

A thin rod of length L carries charge Q spread uniformly along it. (a) Show that the field at a point on the rod's axis, a distance d beyond one end, is $E = kQ/[d(d + L)]$. (b) Show that, at a distance r from the rod's midpoint along its perpendicular bisector, $E = \frac{2kQ}{r\sqrt{L^2 + 4r^2}}$. (c) Show that both results reduce to kQ/r^2 far from the rod, and that (b) reduces to the infinite-line result when $L \rightarrow \infty$.

P13.14

Derive the on-axis field of a uniformly charged disc, $E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right)$, by summing the fields of thin rings. Show that it tends to $\sigma/2\epsilon_0$ close to the disc ($x \ll R$), and to kQ/x^2 far away ($x \gg R$).

P13.15

A thin rod bent into a semicircle of radius R carries charge Q spread uniformly. Find the electric field at the centre of the semicircle.

P13.16

A ring of radius R carries a positive charge Q . An electron (mass m , charge $-e$) is placed on the axis close to the centre and released. (a) Show that for small displacements $x \ll R$ the electron performs SHM, with $\omega^2 = \frac{kQe}{mR^3}$. (b) Find the oscillation frequency for $Q = 1.0$ nC and $R = 1.0$ cm.

P13.17

(a) Starting from the fields of the two point charges, derive the field on the axis of a dipole at distance $r \gg d$: $E = 2kp/r^3$. (b) Estimate the field 1.0 nm from a water molecule along its dipole axis, and the force this field exerts on a sodium ion ($+e$). Compare with the force between the same ion and a single electron 1.0 nm away.

P13.18

Millikan's oil-drop experiment. An oil drop (density 900 kg/m^3) falls through air (viscosity $1.8 \times 10^{-5} \text{ Pa s}$) at a terminal speed of $1.09 \times 10^{-4} \text{ m/s}$ when no field is applied. (a) Using Stokes' law, find the drop's radius and mass. Ignore the buoyancy of air. (b) A vertical field of $7.68 \times 10^4 \text{ N/C}$ holds the drop stationary. Find its charge, as a multiple of e . (c) Explain how Millikan's measurements on many drops showed that charge is quantised.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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